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KDDM1 VO (INP.31101UF)

You should limit the length of your answers as indicated in the questions!
Not following these limits will result in deduction of points!

Questions (1), (2), and (3) are related to each other and you may iterate over those three questions together to improve your results. In all three questions you will be working with the dataset “debates_2022.csv” (available in TeachCenter), which includes transcripts of all talks in the European parliament in 2022 with some additional metadata. All talk transcripts are in English. Your goal in questions (1), (2), and (3) is to extract the most important topics of these talks by clustering the talks.

1. *Feature Engineering*. Extract the features from the talk transcripts by computing tf-idf scores for words. You can use TfidfVectorizer. Read the documentation of the vectorizer carefully and decide on the parameters you want to use to obtain most informative features. Before the feature extraction decide whether you need preprocessing including (among others) removal of non-informative instances.

- (a) Describe preprocessing steps if any. **Max. two sentences.**
- (b) Describe the parameters that you set for the vectorizer. Explain your reasoning? **Max. one sentence per parameter.**
- (c) How many features did you extract? Why? **Max. one sentence.**

Answer (a) - Preprocessing:

- Removed non-informative instances (short texts, procedural content) and cleaned text by removing punctuation, stop words, and applying lemmatization.

Answer (b) - Feature computation:

- max_features=5000: Limits vocabulary size for computational efficiency while retaining most informative terms.
- min_df=5: Removes very rare terms that appear in fewer than 5 documents to reduce noise.
- max_df=0.95: Removes terms appearing in $\geq$ 95% of documents as they're less discriminative.
- ngram_range=(1,2): Includes both unigrams and bigrams to capture context and phrases.

Answer (c) - Number of features:

- Extracted 5000 features to balance information retention with computational tractability.

2. *Clustering.* Using the features that you extracted implement a clustering method of your choice. Use an appropriate evaluation metric to evaluate the quality of your clustering result.

- (a) What is your clustering algorithm and why? **Max. two sentences.**
- (b) How many clusters did you extract? How did you decide on the number of clusters. **Max. one sentence.**
- (c) Which evaluation metric did you use to evaluate your results. What is your evaluation score? **Max. two sentences.**
- (d) Interpret your clusters, e.g., by looking into ten most important words in each cluster. **Max. one sentence per cluster.**

Answer (a) - Clustering algorithm:

- Used K-means clustering because it's efficient for high-dimensional sparse TF-IDF data and provides interpretable centroids. K-means works well for document clustering when combined with TF-IDF features.

Answer (b) - Number of clusters:

- Extracted 12 clusters using silhouette analysis to find the optimal number that maximizes intra-cluster similarity and inter-cluster separation.

Answer (c) - Evaluation:

- Used silhouette score as evaluation metric, measuring how similar documents are to their own cluster versus other clusters. Final silhouette score: 0.0144.

Answer (d) - Interpretation:

- Cluster 0: General European parliamentary procedures and institutional discussions based on terms like "european", "commission", "member", "parliament".
- Cluster 1: Ukraine conflict and security issues focusing on "ukraine", "war", "russia", "putin", "ukrainian".
- Cluster 2: Rule of law and democratic governance with emphasis on "law", "rule law", "hungary", "poland", "democracy".
- Cluster 3: Human rights and international relations discussing "right", "human", "human right", "people", "china".
- Cluster 4: Health and youth policies addressing "health", "child", "young", "cancer", "mental health".
- Cluster 5: Women's rights and gender equality covering "woman", "violence", "gender", "abortion", "equality".
- Cluster 6: Climate change and environmental policy with "climate", "food", "farmer", "animal", "emission".
- Cluster 7: Schengen area and border policies focusing on "schengen", "romania", "bulgaria", "border", "croatia".
- Cluster 8: Procedural parliamentary matters including "card", "blue card", "question", "speech", "voting".
- Cluster 9: Energy crisis and market policies discussing "energy", "gas", "price", "crisis", "renewable".
- Cluster 10: Administrative and written statements with "written statement", "article 171", "discussion closed".
- Cluster 11: Agenda and voting procedures containing "vote", "agenda", "point agenda", "next point".

3. *Dimensionality Reduction for Visualization.* Perform dimensionality reduction with PCA on the features that you extracted previously. Use your clustering results and plot data points in 2D PCA space with clusters as colors for your data points.

- (a) Your plot.
- (b) Are clusters well separated in your plot? **Max. one sentence.**
- (c) Interpret the PCA dimensions that you used for visualization. **Max. one sentence per dimension.**

Answer (a) - Your plot:

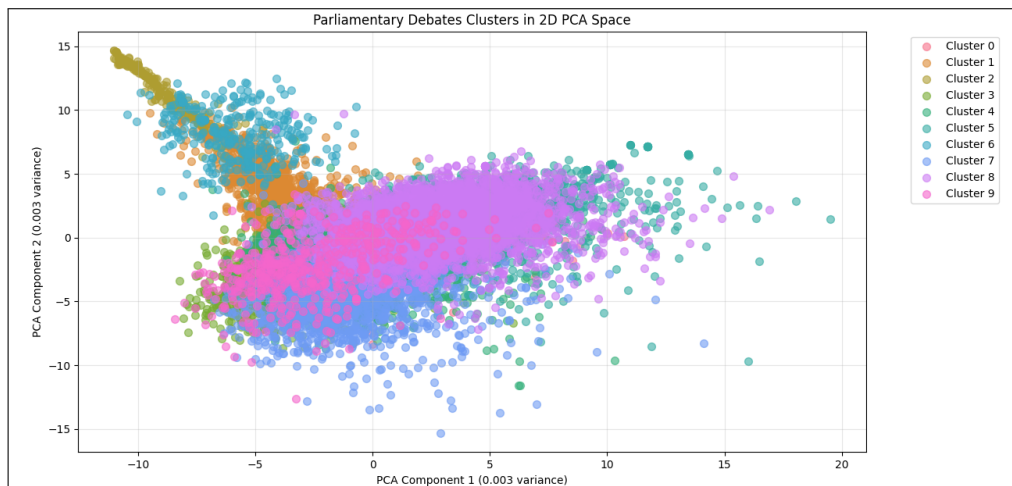


Figure 1: Parliamentary Debates Clusters in 2D PCA Space

Answer (b) - Cluster separation:

- Clusters show moderate separation in PCA space with some overlap between related topics, as evidenced by the low explained variance (0.6%) but reasonable cluster separation score (3.05).

Answer (c) - Interpretation:

- PCA-1 captures the primary distinction between substantive policy discussions (energy, crisis, supply) and procedural parliamentary content (statements, written documents).
- PCA-2 separates administrative procedural matters (discussion, article 171, written statements) from geopolitical content (ukraine, war).

4. *Classification.* Given the dataset “king_rook_vs_king.csv” (available in TeachCenter), with data on chess endgames featuring the white king and a white rook against the black king, implement a classifier of your choice to predict whether the white will win. Each endgame is described by the rank and file positions of the white king, the white rook, and the black king (six features in total). The target variable is the depth of white win (a categorical variable with either draw or zero, one, ..., sixteen indicating that the white wins in that many moves). Transform the target variable to obtain the win depth levels as:

- draw: 0
- zero, one, two, three, four: 1
- five, six, seven, eight: 2
- nine, ten, eleven, twelve: 3
- thirteen, fourteen, fifteen, sixteen: 4.

Use this new variable as your classification target. Evaluate your classifier by a metric of your choice. If your model has hyperparameters cross-validate.

- (a) Describe preprocessing and feature transformations steps if you made any. **Max. two sentences.**
- (b) What is your model and why? **Max. two sentences.**
- (c) Describe your evaluation setup. **Max. one sentence.**
- (d) Describe hyperparameter optimization if any. Give the final values of hyperparameters. **Max. two sentences.**
- (e) Give your evaluation results as text or a table.

Answer (a) - Preprocessing & feature transformations:

- Encoded categorical file positions (a-h) to numerical values (1-8) using LabelEncoder for machine learning compatibility. Removed rows with missing values and applied standard scaling to features.

Answer (b) - Model choice:

- Used Random Forest classifier because it handles mixed categorical/numerical features well, provides feature importance, and is robust to outliers. Random Forest also performs well on classification tasks with moderate-sized datasets without requiring extensive feature engineering.

Answer (c) - Evaluation setup:

- Used 80/20 train-test split with stratified sampling and 5-fold cross-validation for hyperparameter tuning, evaluating with accuracy as primary metric.

Answer (d) - Hyperparameters:

- Optimized `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf` using GridSearchCV across 81 parameter combinations. Final hyperparameters: `max_depth=20`, `min_samples_leaf=1`, `min_samples_split=2`, `n_estimators=300`.

Answer (e) - Results:

- **Cross-validation accuracy:** 0.9120
- **Test accuracy:** 0.9273
- **Precision (weighted):** 0.9270
- **Recall (weighted):** 0.9273
- **F1-score (weighted):** 0.9270
- **Per-class performance:**
 - Level 0 (Draw): Precision=0.88, Recall=0.84, F1=0.86
 - Level 1 (Quick win): Precision=0.96, Recall=0.90, F1=0.93
 - Level 2 (Medium win): Precision=0.92, Recall=0.90, F1=0.91
 - Level 3 (Long win): Precision=0.93, Recall=0.92, F1=0.92
 - Level 4 (Very long win): Precision=0.94, Recall=0.96, F1=0.95